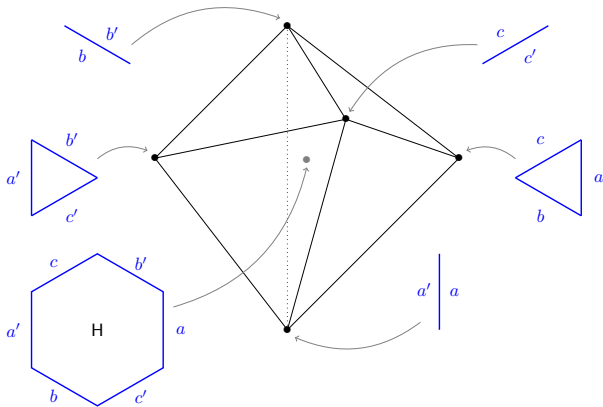


Minkowski indecomposability of polytopes

Germain Poullot with Arnau Padrol
ArXiv: 2512.05307



10 December 2025

- 1 What is a polytope?
 - Two definitions of polytopes
 - Faces and normal fans

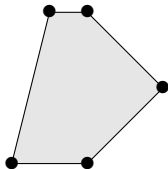
- 2 Deformations (a.k.a. Minkowski summands)
 - Minkowski decomposition
 - Cone of deformations

- 3 Graph of edge dependencies
 - Triangle, parallelograms
 - Graph of edge dependencies
 - Applications

What is a polytope?

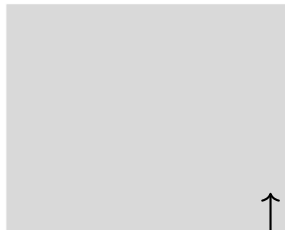
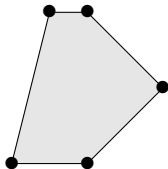
Definition

Polytope: convex hull of finitely many points in $\mathbb{R}^n$



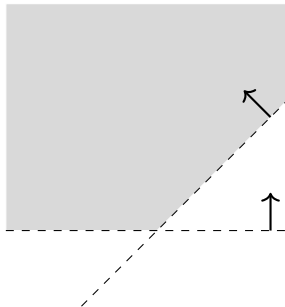
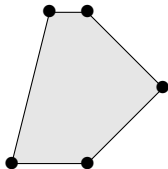
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Polytope: convex hull of finitely many points in $\mathbb{R}^n$
bounded intersection of finitely many half-spaces in $\mathbb{R}^n$



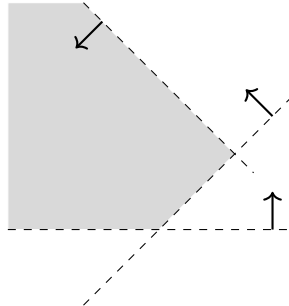
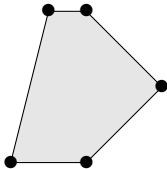
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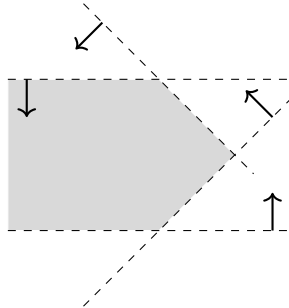
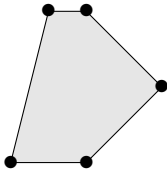
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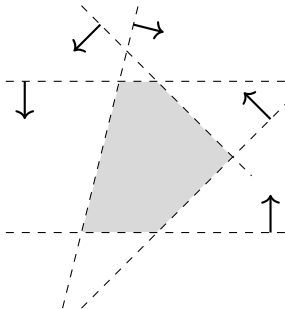
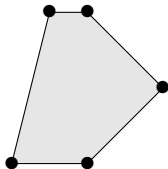
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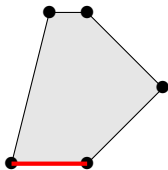
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Definition

Face: $P^c := \{ \mathbf{x} \in \mathbb{R}^n ; \langle \mathbf{x}, \mathbf{c} \rangle = \max_{\mathbf{y} \in P} \langle \mathbf{y}, \mathbf{c} \rangle \}$

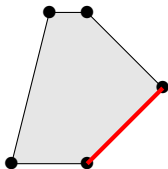


P



Definition

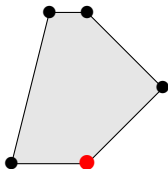
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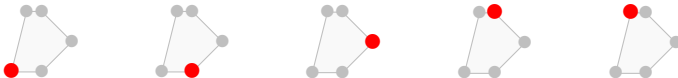


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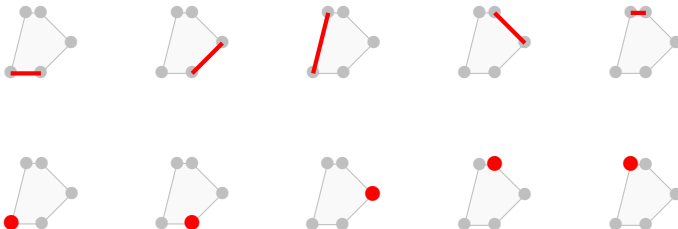
Face lattice

Face lattice: poset of inclusions of faces



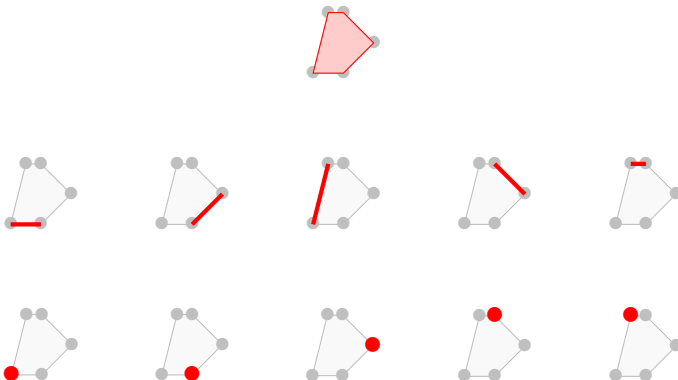
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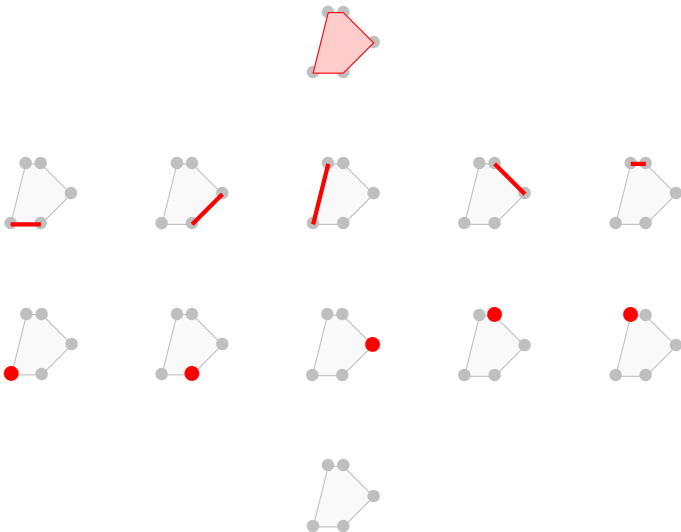
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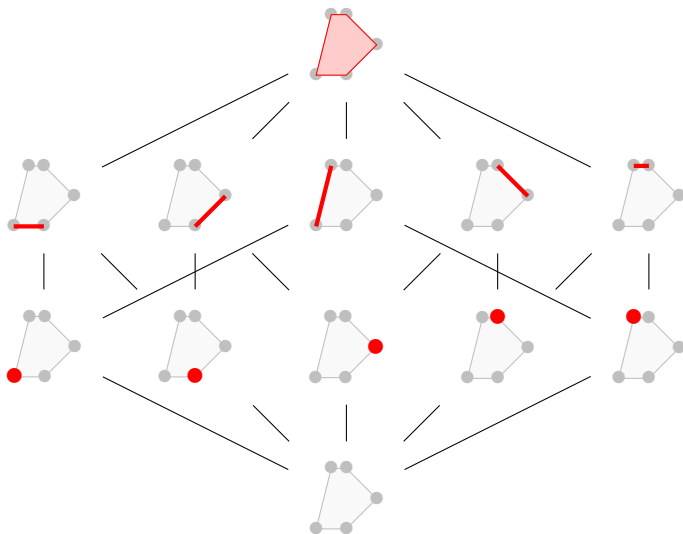
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Deformations (a.k.a. Minkowski summands)

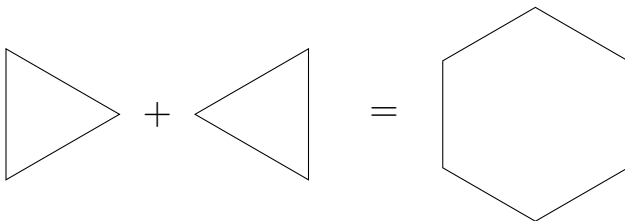
Minkowski sum

Definition

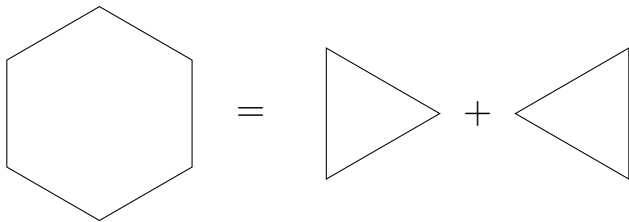
P, Q polytopes. *Minkowski sum*:

$$P + Q = \{ \mathbf{p} + \mathbf{q} \ ; \ \mathbf{p} \in P, \ \mathbf{q} \in Q \}$$

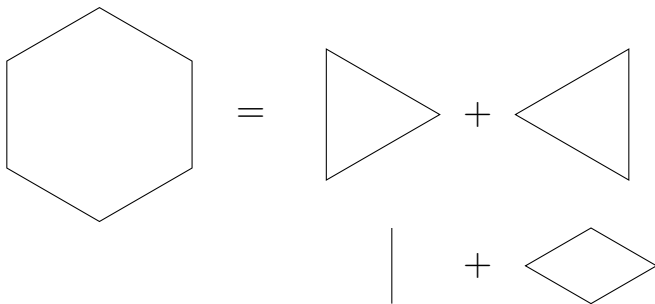
N.B. $\text{Vert}(P + Q) \subseteq \text{Vert}(P) + \text{Vert}(Q)$



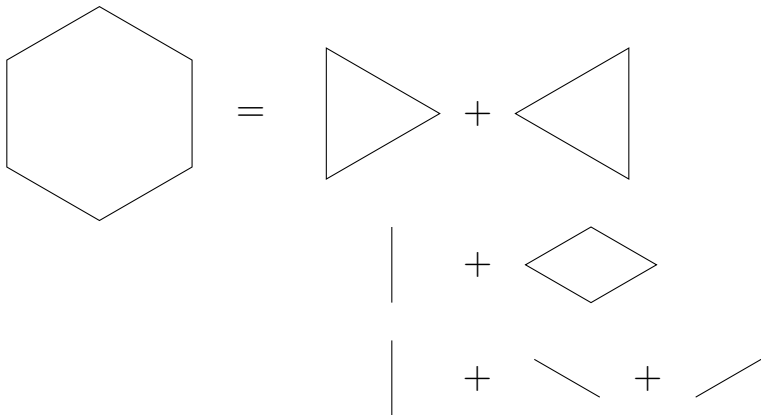
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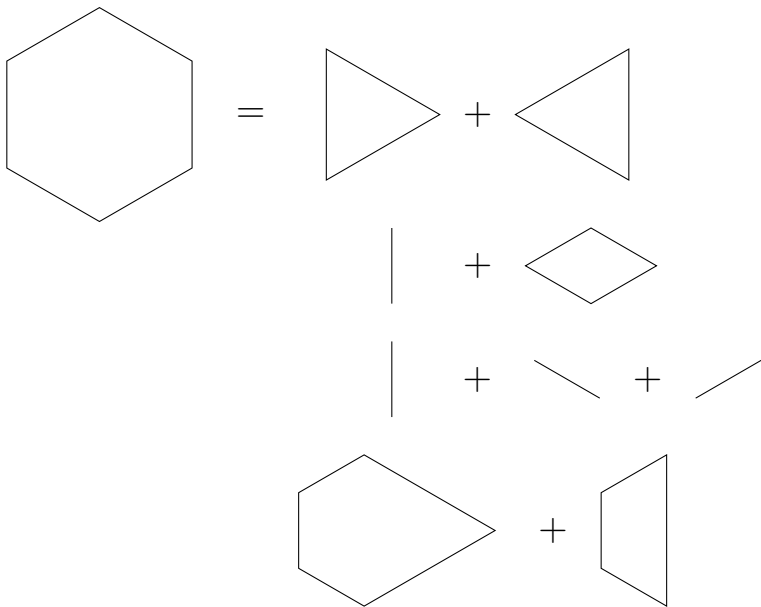
Minkowski sum



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Definition

Q is a *Minkowski summand*, a.k.a. *deformation*, of P when there exists R and $\lambda > 0$ with:

$$\lambda P = Q + R$$

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What is the best way to write P as a Minkowski sum ?

- With the fewest number of (indecomposable) summands ?
- With the (indecomposable) summands of smallest dimension ?
- Respecting some symmetries ?
- ...

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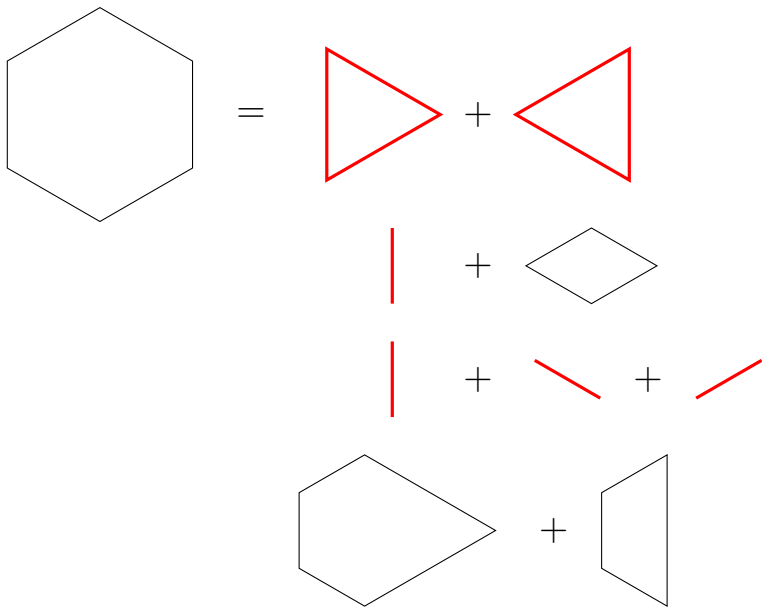
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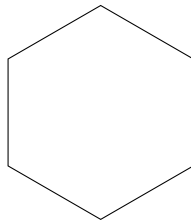
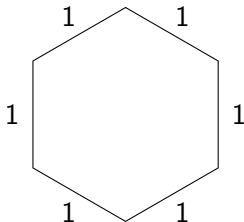
$\implies$ What is the structure of $\mathbb{DC}(P)$?

Minkowski summands



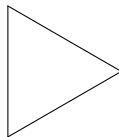
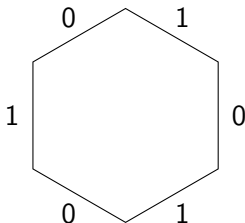
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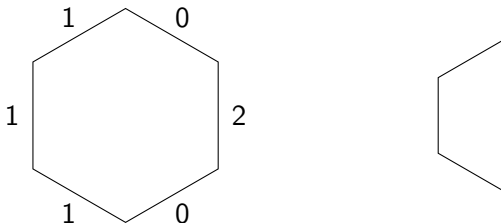
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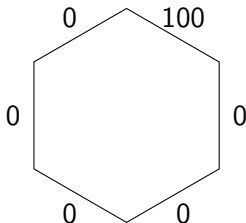
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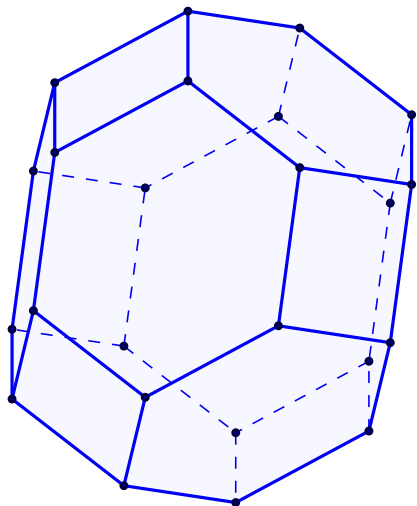


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Deformations of 3-dim permutahedron



Permutahedron Π_4

Sequence of deformations of Π_4

Edge-length deformation cone

Theorem

Q deformation of $P \Leftrightarrow$ same edge-directions, but different lengths

Definition

Edge-length deformation cone: $\mathbb{DC}(P) = \{Q ; Q \text{ same edge-dir } P\}$

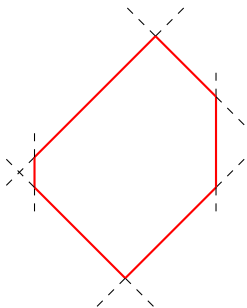
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edge-length vector:

$$\ell = (\ell_e)_{e \text{ edge}}$$

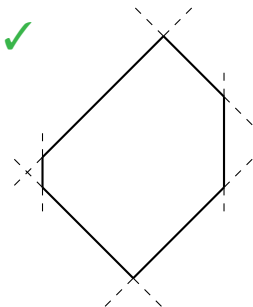
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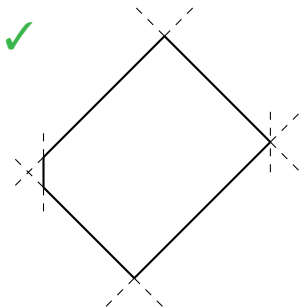
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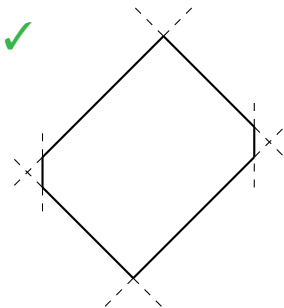
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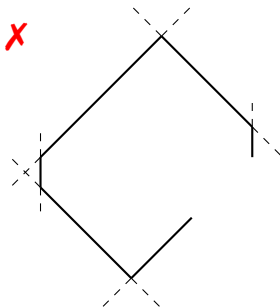
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Cycle equations:

linear equations on ℓ

$$\ell_e \geq 0 \text{ for all } e \text{ edge}$$

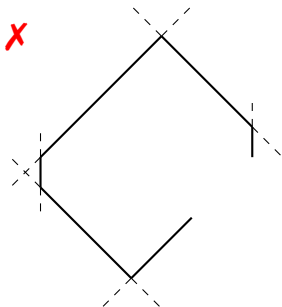
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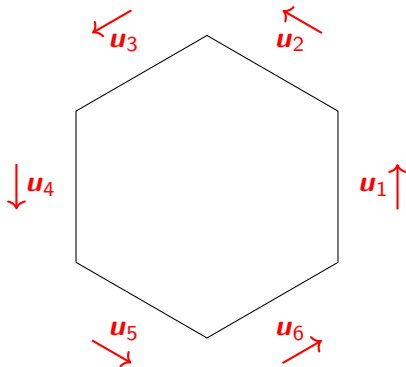
Cycle equations:

linear equations on ℓ

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P_ℓ = start at a vertex, find the coordinates of the other vertices from the graph of P and ℓ

Cycle equations



For F a 2-dim face of P :

$$\sum_e u_e = \mathbf{0} \quad , \quad u_e \text{ unit vector}$$

hence

$$\sum_e \ell_e u_e = \mathbf{0}$$

Summary on $\mathbb{DC}(P)$

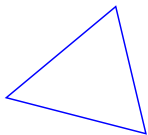
	$\mathbb{DC}(P)$	
Q	ℓ	h
Minkowski summands	edge-lengths	heights on rays
$Q_1 + Q_2$	$\ell_1 + \ell_2$	$h_1 + h_2$
Dilation λQ	$\lambda \ell$	λh
Translations	Has been fixed	Lineal
<i>complicated</i>	edge directions Cycle equations V -description	normal fan $\mathcal{N}_P$ Wall-crossing ineq. H -description
Polytope algebra	Weight algebra	Polynomial algebra

$\mathbb{DC}(P)$ is a ray = P Minkowski indecomposable

$\mathbb{DC}(P)$ is simplicial cone = P has **unique** Minkowski decomposition

Graph of edge dependencies

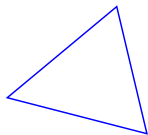
Cycle equation of a triangle, quadrilateral



Triangle:

3 variables (= lengths of the edges)

Cycle equation of a triangle, quadrilateral

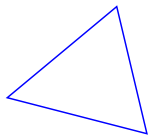


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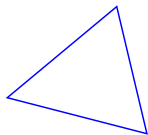
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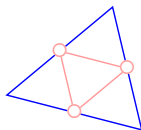
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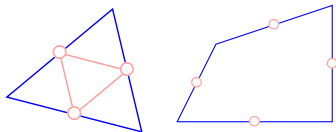
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$\Rightarrow$ If I know the length of 1 edge, I know the length of the others.

Cycle equation of a triangle, quadrilateral



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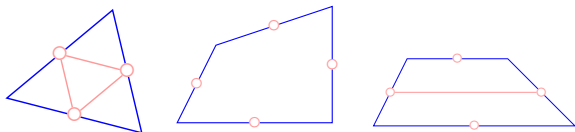
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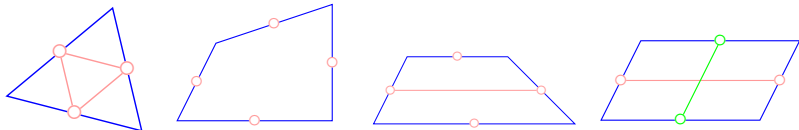
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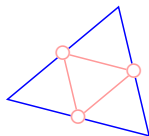
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Graph of edge dependencies

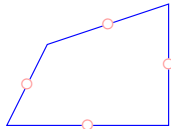
Graph of edge dependencies $ED(P)$:

node: edges of P

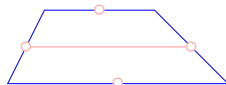
arcs: link two *dependent* edges, i.e. I can deduce the length of one from the length of the other, using the cycle equations



(a) Triangle



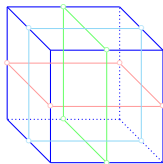
(b) Quadrilateral



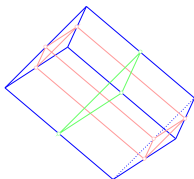
(c) Trapezoid



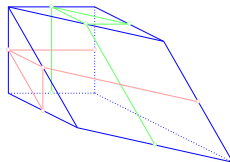
(d) Parallelogram



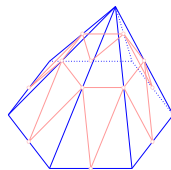
(e) Cube



(f) Prism



(g) Hemicube



(h) Pyramid

Problem 1:

How to find edges of $ED(P)$.

The two problems

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A Gale's criterion, '54

All 2-faces of P are triangles $\Rightarrow P$ indecomposable.

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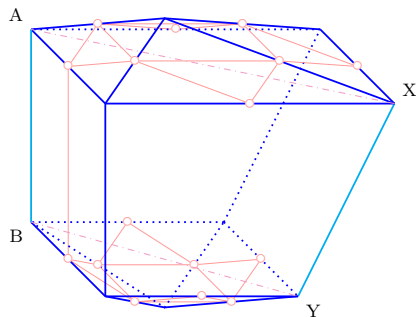
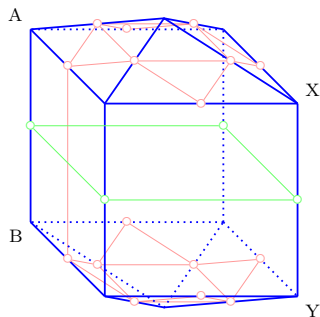
"triangles" $\rightarrow$ edges in $ED(P)$

"all 2-faces" $\rightarrow$ enough edges in $ED(P)$

Problem 1: creating edges

In $ED(P)$, two node e, f ($\in$ edges of P) are linked if, e.g.:

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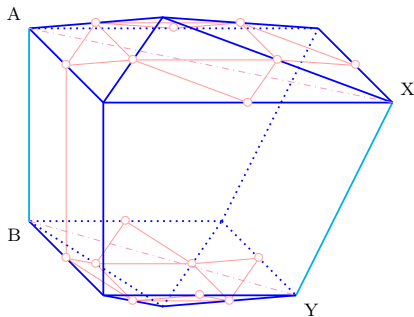
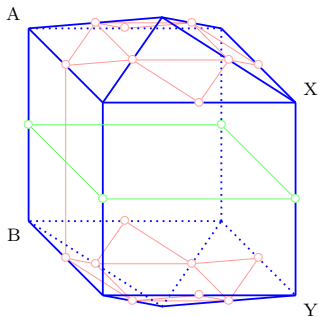


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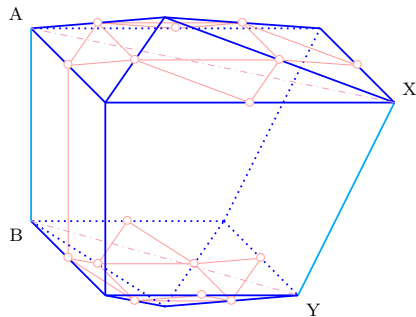
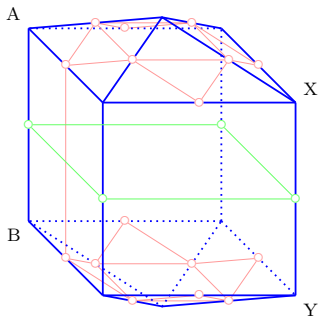
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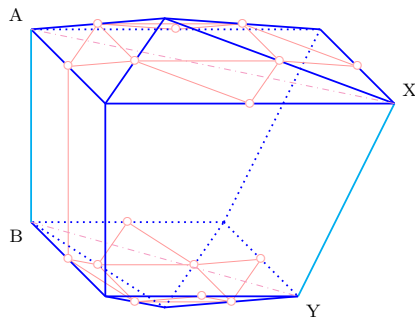
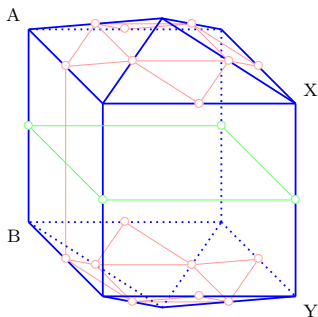
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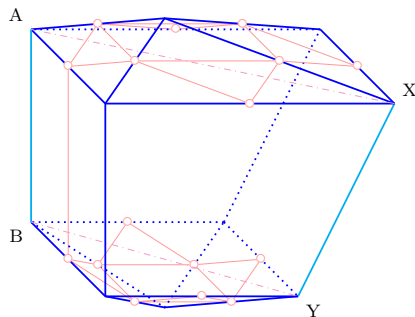
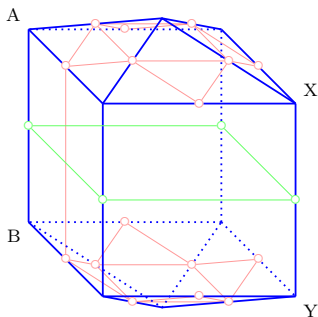
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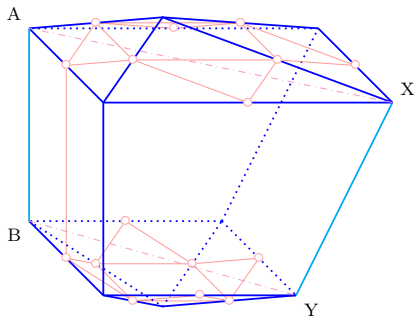
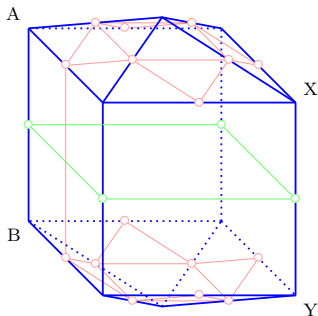


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- ... (there is a nicer way to write these properties)

+ implicit edges



Problem 2: consequences of $ED(P)$

Theorem (Padrol–P. '25)

If $ED(P)$ is connected, then P is indecomposable

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If there exists a set S of dependent vertices such that every facet contains a vertex of S , then P is indecomposable.

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Coro.: Gale, Kallay, Shepard, McMullen, Yost–Przesławski criteria...

Corollary (Padrol–P. '25)

If there is $S \subseteq V(P)$, and $X_1, \dots, X_r$ dependent sets of edges with

- any two vertices of S are connected via $\bigcup_i X_i$, and*
- every facet of P contains a vertex of S ,*

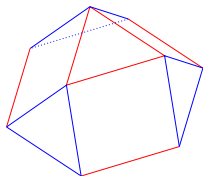
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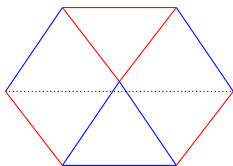
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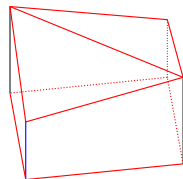
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(a) Triang. cupola

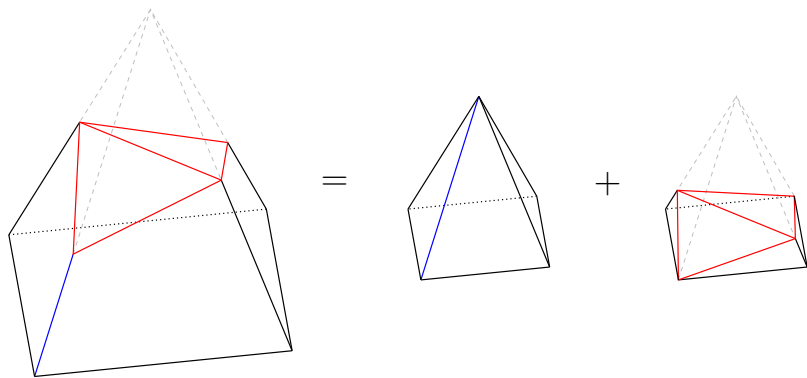


(b) Sum of 2 triangles



(c) Chiseled cube

$\dim \mathbb{DC}(P) = 2$, only one Minkowski decomposition (in 2 terms)



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P, Q are in *parallelogramic position* if no edge of P is in a plane spanned by the 2-faces of Q

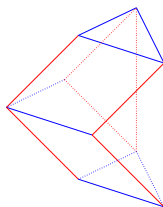
Theorem (Padrol–P. '25)

P, Q in parallelogramic position: $\mathbb{DC}(P + Q) \simeq \mathbb{DC}(P) \times \mathbb{DC}(Q)$.

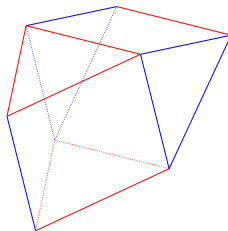
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(a) Diminished trapezohedron



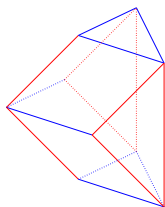
(b) Gyrobifastigium

Besides

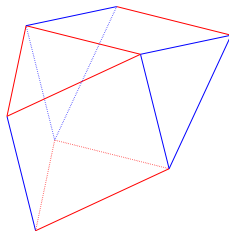
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Corollary (Padrol–P. '25)

$\mathbb{DC}(P \times Q) \simeq \mathbb{DC}(P) \times \mathbb{DC}(Q)$

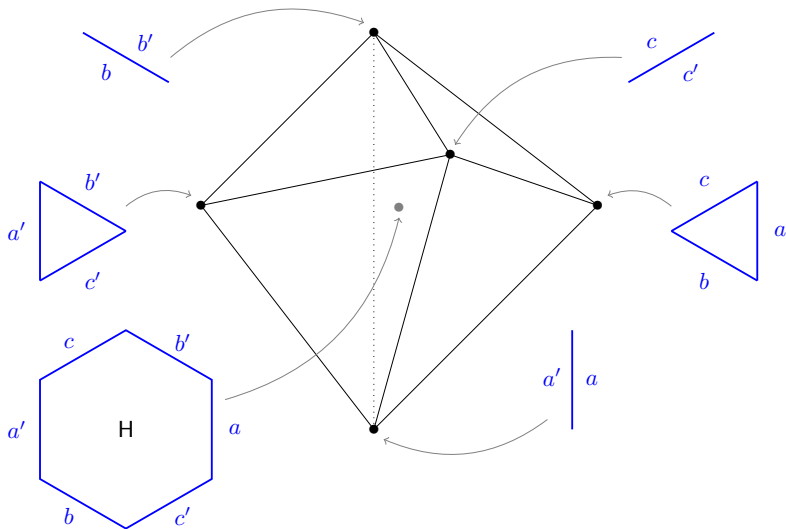
Other applications:

- new indecomposable generalized permutahedra
- “autonomous and dependent” parts of $ED(P)$ give rays of $\mathbb{DC}(P)$
- you can stack/delete vertices on polytopes, study indecomposability
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Be careful, $ED(P)$ is not always sufficient for recovering $\mathbb{DC}(P)$.



Merci !

Thank you!